

MATHEMATICS

Chapter 7: Integrals

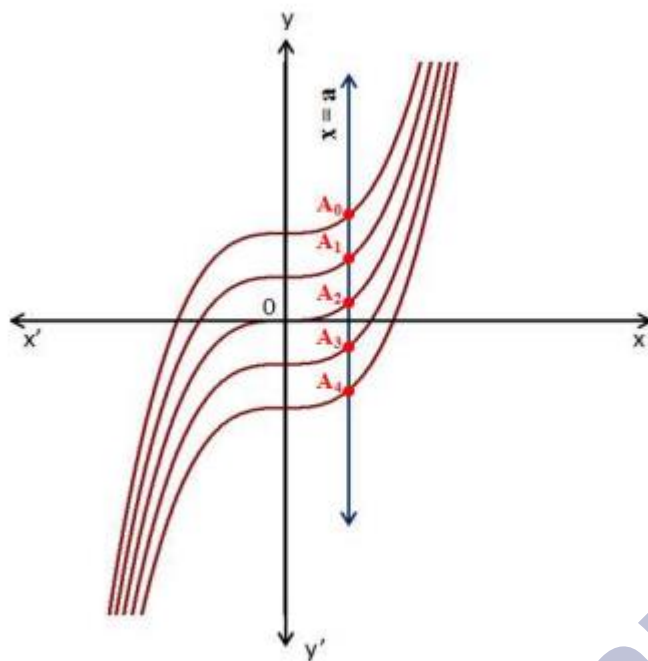


INTEGRALS

Top Concepts

1. Integration is the inverse process of differentiation. The process of finding the function from its primitive is known as integration or antidifferentiation.
2. The problem of finding a function whenever its derivative is given leads to indefinite form of integrals.
3. The problem of finding the area bounded by the graph of a function under certain conditions leads to a definite form of integrals.
4. Indefinite and definite integrals together constitute **Integral Calculus**.
5. Indefinite integral $\int f(x)dx = F(x) + C$, where $F(x)$ is the antiderivative of $f(x)$.
6. Functions with same derivatives differ by a constant.
7. $\int f(x)dx$ means integral of f with respect to x , $f(x)$ is the integrand, x is the variable of integration and C is the constant of integration.
8. Geometrically indefinite integral is the collection of family of curves, each of which can be obtained by translating one of the curves parallel to itself.

Family of curves representing the integral of $3x^2$



$\int f(x)dx = F(x) + C$, represents a family of curves where different values of C correspond to different members of the family, and these members are obtained by shifting any one of the curves parallel to itself.

9. Properties of antiderivatives

$$\int [f(x) + g(x)]dx = \int f(x)dx + \int g(x)dx$$

$$\int kf(x)dx = k \int f(x)dx \text{ for any real number } k$$

$$\int [k_1f_1(x) + k_2f_2(x) + \dots + k_nf_n(x)]dx = k_1 \int f_1(x)dx + k_2 \int f_2(x)dx + \dots + k_n \int f_n(x)dx$$

where $k_1, k_2, \dots, k_n$ are real numbers and $f_1, f_2, \dots, f_n$ are real functions.

10. Two indefinite integrals with the same derivative lead to the same family of curves and so they are equivalent.

11. Comparison between differentiation and integration

1. Both are operations on functions.
2. Both satisfy the property of linearity.
3. All functions are not differentiable and all functions are not integrable.
4. The derivative of a function is a unique function, but the integral of a function is not.
5. When a polynomial function P is differentiated, the result is a polynomial whose degree is 1 less than the degree of P . When a polynomial function P is integrated, the result is a polynomial whose degree is 1 more than that of P .

6. The derivative is defined at a point P and the integral of a function is defined over an interval.
7. Geometrical meaning: The derivative of a function represents the slope of the tangent to the corresponding curve at a point. The indefinite integral of a function represents a family of curves placed parallel to each other having parallel tangents at the points of intersection of the family with the lines perpendicular to the axis.
8. The derivative is used for finding some physical quantities such as the velocity of a moving particle when the distance traversed at any time t is known. Similarly, the integral is used in calculating the distance traversed when the velocity at time t is known.
9. Differentiation and integration, both are processes involving limits.
10. By knowing one antiderivative of function f, an infinite number of antiderivatives can be obtained.
12. Integration can be done by using many methods. Prominent among them are
 1. Integration by substitution
 2. Integration using partial fractions
 3. Integration by parts
 4. Integration using trigonometric identities.
13. A change in the variable of integration often reduces an integral to one of the fundamental integrals. Some standard substitutions are

$x^2 + a^2$; substitute $x = a \tan \theta$

$\sqrt{x^2 - a^2}$; substitute $x = a \sec \theta$

$\sqrt{a^2 - x^2}$; substitute $x = a \sin \theta$ or $a \cos \theta$
14. A function of the form $\frac{P(x)}{Q(x)}$ is known as a rational function. Rational functions can be integrated using partial fractions.
15. **Partial fraction decomposition or partial fraction expansion** is used to reduce the degree of either the numerator or the denominator of a rational function.
16. **Integration using partial fractions**

A rational function $\frac{P(x)}{Q(x)}$ can be expressed as the sum of partial fractions if $\frac{P(x)}{Q(x)}$. This takes any of the forms:

- $\frac{px+q}{(x-a)(x-b)} = \frac{A}{x-a} + \frac{B}{x-b}, a \neq b$
- $\frac{px+q}{(x-a)^2} = \frac{A}{x-a} + \frac{B}{(x-a)^2}$
- $\frac{px^2+qx+r}{(x-a)(x-b)(x-c)} = \frac{A}{x-a} + \frac{B}{x-b} + \frac{C}{x-c}$
- $\frac{px^2+qx+r}{(x-a)^2(x-b)} = \frac{A}{x-a} + \frac{B}{(x-a)^2} + \frac{C}{x-b}$
- $\frac{px^2+qx+r}{(x-a)(x^2+bx+c)} = \frac{A}{x-a} + \frac{Bx+C}{x^2+bx+c}$

where $x^2 + bx + c$ cannot be factorised further.

17. To find the integral of the product of two functions, integration by parts is used. I and II functions are chosen using the ILATE rule:

I - inverse trigonometric

L - logarithmic

A - algebraic

T - trigonometric

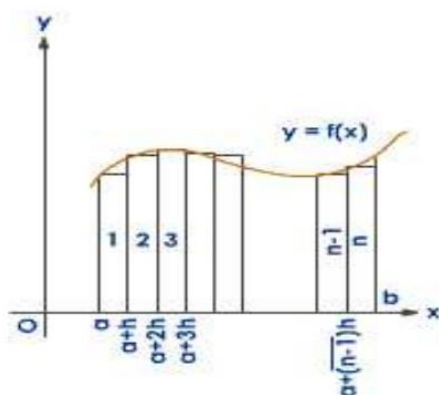
E - exponential is used to identify the first function.

18. Integration by parts

Integral of the product of two functions = (first function) $\times$ (integral of the second function) - integral of [(differential coefficient of the first function) $\times$ (integral of the second function)].

$$\int f_1(x) \cdot f_2(x) dx = f_1(x) \int f_2(x) dx - \int \left[\frac{d}{dx} f_1(x) \cdot \int f_2(x) dx \right] dx, \text{ where } f_1 \text{ and } f_2 \text{ are functions of } x.$$

19. Definite integral $\int_a^b f(x) dx$ of the function $f(x)$ from limits a to b represents the area enclosed by the graph of the function $f(x)$, the x -axis and the vertical markers $x = 'a'$ and $x = 'b'$.



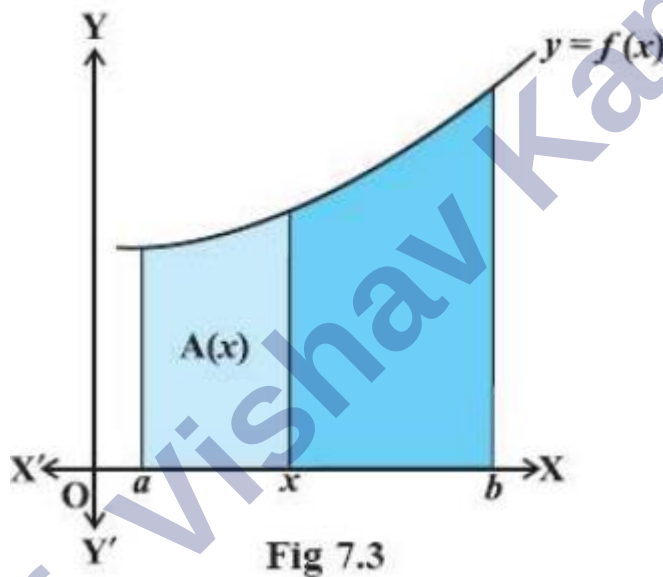
20. **Definite integral as the limit of a sum:** The process of evaluating a definite integral by using the definition is called integration as the limit of a sum or integration from first principles.

21. Method of evaluating $\int_a^b f(x)dx$

- Calculate antiderivative $F(x)$
- Calculate $F(b) - F(a)$

22. Area function

$$A(x) = \int_a^x f(x)dx, \text{ if } x \text{ is a point in } [a, b].$$



23. Fundamental Theorem of Integral Calculus

- First fundamental theorem** of integral calculus: If area function, $A(x) = \int_a^x f(x)dx$ for all $x \geq a$, and f is continuous on $[a, b]$. Then $A'(x) = f(x)$ for all $x \in [a, b]$
- Second fundamental theorem** of integral calculus: Let f be a continuous function of x in the closed interval $[a, b]$ and let F be antiderivative of $\frac{d}{dx} f(x) = f(x)$ for all x in domain of f , then

$$\int_a^b f(x)dx = [F(x) + C]_a^b = F(b) - F(a)$$

Top Formulae

1. Some Standard Integrals

- $\int x^n dx = \frac{x^{n+1}}{n+1} + C, n \neq -1$
- $\int dx = x + C$
- $\int \cos x dx = \sin x + C$
- $\int \sin x dx = -\cos x + C$
- $\int \sec^2 x dx = \tan x + C$
- $\int \operatorname{cosec}^2 x dx = -\cot x + C$
- $\int \sec x \tan x dx = \sec x + C$
- $\int \operatorname{cosec} x \cot x dx = -\operatorname{cosec} x + C$
- $\int \frac{dx}{\sqrt{1-x^2}} = \sin^{-1} x + C$
- $\int -\frac{dx}{\sqrt{1-x^2}} = \cos^{-1} x + C$
- $\int \frac{dx}{1+x^2} = \tan^{-1} x + C$
- $\int \frac{dx}{1+x^2} = -\cot^{-1} x + C$
- $\int \frac{dx}{x\sqrt{x^2-1}} = \sec^{-1} x + C$
- $\int \frac{dx}{x\sqrt{x^2-1}} = -\operatorname{cosec}^{-1} x + C$
- $\int e^x dx = e^x + C$
- $\int a^x dx = \frac{a^x}{\log a} + C$

- $\int \frac{1}{x} dx = \log|x| + C$
- $\int \tan x dx = \log|\sec x| + C$
- $\int \cot x dx = \log|\sin x| + C$
- $\int \sec x dx = \log|\sec x + \tan x| + C$
- $\int \operatorname{cosec} x dx = \log|\operatorname{cosec} x - \cot x| + C$
- $\int (ax + b)^n dx = \frac{(ax + b)^{n+1}}{a(n+1)} + C, n \neq -1$
- $\int \frac{1}{ax + b} dx = \frac{1}{a} \log|ax + b| + C$
- $\int e^{ax+b} dx = \frac{1}{a} e^{ax+b} + C$
- $\int a^{bx+c} dx = \frac{1}{b} \cdot \frac{a^{bx+c}}{\log a} + C, a > 0, a \neq 1$
- $\int \sin(ax + b) dx = -\frac{1}{a} \cos(ax + b) + C$
- $\int \cos(ax + b) dx = \frac{1}{a} \sin(ax + b) + C$
- $\int \tan(ax + b) dx = \frac{1}{a} \log|\sec(ax + b)| + C$
- $\int \cot(ax + b) dx = \frac{1}{a} \log|\sin(ax + b)| + C$
- $\int \sec^2(ax + b) dx = \frac{1}{a} \tan(ax + b) + C$
- $\int \operatorname{cosec}^2(ax + b) dx = -\frac{1}{a} \cot(ax + b) + C$
- $\int \sec(ax + b) \tan(ax + b) dx = \frac{1}{a} \sec(ax + b) + C$
- $\int \operatorname{cosec}(ax + b) \cot(ax + b) dx = -\frac{1}{a} \operatorname{cosec}(ax + b) + C$
- $\int \sec(ax + b) dx = \frac{1}{a} \log|\sec(ax + b) + \tan(ax + b)| + C$
- $\int \operatorname{cosec}(ax + b) dx = \frac{1}{a} \log|\operatorname{cosec}(ax + b) - \cot(ax + b)| + C$

2. Integral of some special functions

- $\int \frac{dx}{x^2 - a^2} = \frac{1}{2a} \log \left| \frac{x-a}{x+a} \right| + C$
- $\int \frac{dx}{a^2 - x^2} = \frac{1}{2a} \log \left| \frac{a+x}{a-x} \right| + C$
- $\int \frac{dx}{x^2 + a^2} = \frac{1}{a} \tan^{-1} \frac{x}{a} + C$
- $\int -\frac{dx}{x^2 + a^2} = \frac{1}{a} \cot^{-1} \frac{x}{a} + C$
- $\int \frac{dx}{\sqrt{x^2 - a^2}} = \log \left| x + \sqrt{x^2 - a^2} \right| + C$
- $\int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1} \frac{x}{a} + C$
- $\int -\frac{dx}{\sqrt{a^2 - x^2}} = \cos^{-1} \frac{x}{a} + C$
- $\int \frac{dx}{x\sqrt{x^2 - a^2}} = \frac{1}{a} \sec^{-1} \frac{x}{a} + C$
- $\int -\frac{dx}{x\sqrt{x^2 - a^2}} = \frac{1}{a} \operatorname{cosec}^{-1} \frac{x}{a} + C$
- $\int \frac{dx}{\sqrt{x^2 + a^2}} = \log \left| x + \sqrt{x^2 + a^2} \right| + C$
- $\int e^{ax} \sin bx dx = \frac{e^{ax}}{a^2 + b^2} (a \sin bx - b \cos bx) + C$
- $\int e^{ax} \cos bx dx = \frac{e^{ax}}{a^2 + b^2} (a \cos bx + b \sin bx) + C$
- $\int \sqrt{a^2 - x^2} dx = \frac{1}{2} x \sqrt{a^2 - x^2} + \frac{1}{2} a^2 \sin^{-1} \frac{x}{a} + C$
- $\int \sqrt{a^2 + x^2} dx = \frac{1}{2} x \sqrt{a^2 + x^2} + \frac{1}{2} a^2 \log \left| x + \sqrt{a^2 + x^2} \right| + C$
- $\int \sqrt{x^2 - a^2} dx = \frac{1}{2} x \sqrt{x^2 - a^2} - \frac{1}{2} a^2 \log \left| x + \sqrt{x^2 - a^2} \right| + C$

3. Integration by parts

$$(i) \int f_1(x) \cdot f_2(x) dx = f_1(x) \int f_2(x) dx - \int \left[\frac{d}{dx} f_1(x) \cdot \int f_2(x) dx \right] dx, \text{ where } f_1 \text{ and } f_2 \text{ are functions of } x.$$

$$(ii) \int e^x (f(x) + f'(x)) dx = e^x f(x) + C$$

4. Integral as the limit of sums:

$$\int_a^b f(x)dx = (b-a) \lim_{n \rightarrow \infty} \frac{1}{n} [f(a) + f(a+h) + \dots + f(a+(n-1)h)] \text{ where } h = \frac{b-a}{n}$$

5. Different methods of integration

1. Evaluation of integrals of the form $\frac{p(x)}{(ax+b)^n}$, $n \in \mathbb{N}$, where $p(x)$ is a polynomial

Steps:

- i. Check whether degree of $p(x) \geq$ or $\leq n$

- ii. If degree of $p(x) < n$, then express $p(x)$ in the form

$$A_0 + A_1(ax+b) + A_2(ax+b)^2 + \dots + A_{n-1}(ax+b)^{n-1}$$

- iii. Write $\frac{p(x)}{(ax+b)^n}$ as $\frac{A_0}{(ax+b)^n} + \frac{A_1}{(ax+b)^{n-1}} + \frac{A_2}{(ax+b)^{n-2}} + \dots + \frac{A_{n-1}}{(ax+b)}$

- iv. Evaluate

$$\int \frac{p(x)}{(ax+b)^n} dx = A_0 \int \frac{1}{(ax+b)^n} dx + A_1 \int \frac{1}{(ax+b)^{n-1}} dx \\ + A_2 \int \frac{1}{(ax+b)^{n-2}} dx + \dots + A_{n-1} \int \frac{1}{(ax+b)} dx$$

- v. If degree of $p(x) \geq n$, then divide $p(x)$ by $(ax+b)^n$ and express

$$\frac{p(x)}{(ax+b)^n} \text{ as } q(x) + \frac{r(x)}{(ax+b)^n}, \text{ where degree of } r(x) \text{ is less than } n$$

- vi. Use steps (ii) and (iii) to evaluate $\int \frac{r(x)}{(ax+b)^n} dx$

2. Evaluation of integrals of the form $\int (ax+b)\sqrt{cx+d} dx$

Step:

- i. Represent $(ax + b)$ in terms of $(cx + d)$ as follows:

$$(ax + b) = A(cx + d) + B$$

- ii. Find A and B by equating coefficients of like powers of x on both sides
 iii. Replace $(ax + b)$ by $A(cx + d) + B$ in the given integral to obtain

$$\begin{aligned}\int (ax + b) \sqrt{cx + d} dx &= \int [A(cx + d) + B] \sqrt{cx + d} dx \\ &= A \int (cx + d)^{\frac{3}{2}} dx + B \int \sqrt{cx + d} dx \\ &= \frac{2A}{5c} (cx + d)^{\frac{5}{2}} + \frac{2B}{5c} (cx + d)^{\frac{3}{2}} + C\end{aligned}$$

3. Evaluation of integrals of the form $\int \frac{(ax+b)}{\sqrt{cx+d}} dx$

Step:

- i. Represent $(ax + b)$ in terms of $(cx + d)$ as follows:

$$(ax + b) = A(cx + d) + B$$

- ii. Find A and B by equating coefficients of like powers of x on both sides
 iii. Replace $(ax + b)$ by $A(cx + d) + B$ in the given integral to obtain

$$\begin{aligned}\int \frac{(ax + b)}{\sqrt{cx + d}} dx &= \int \frac{[A(cx + d) + B]}{\sqrt{cx + d}} dx \\ &= A \int \sqrt{cx + d} dx + B \int \frac{1}{\sqrt{cx + d}} dx \\ &= \frac{2A}{3c} (cx + d)^{\frac{3}{2}} + \frac{2B}{c} (cx + d)^{\frac{1}{2}} + C\end{aligned}$$

4. Evaluation of integrals of the form $\int x dx, \int \cos^m x dx, \text{ where } m \leq 4$

Let us express $\sin^m x$ and $\cos^m x$ in terms of sines and cosines of multiples of x by using the following identities:

- i. $\sin^2 x = \frac{1 - \cos 2x}{2}$
- ii. $\cos^2 x = \frac{1 + \cos 2x}{2}$
- iii. $\sin 3x = 3 \sin x - 4 \sin^3 x$
- iv. $\cos 3x = 4 \cos^3 x - 3 \cos x$

5. Evaluation of integrals of the form

$$\int \sin mx \cdot \cos nx dx, \int \sin mx \cdot \sin nx dx, \int \cos mx \cdot \cos nx dx$$

Let us use the following identities:

- i. $2 \sin A \cos B = \sin(A + B) + \sin(A - B)$
- ii. $2 \cos A \cos B = \cos(A + B) + \cos(A - B)$
- iii. $2 \cos A \sin B = \sin(A + B) - \sin(A - B)$
- iv. $2 \sin A \sin B = \cos(A - B) - \cos(A + B)$

6. Evaluation of integrals of the form $\int \frac{f'(x)}{f(x)} dx$

$$\int \frac{f'(x)}{f(x)} dx = \log|f(x)| + C$$

7. Evaluation of integrals of the form

$$\int (ax + b)^n p(x) dx, \int \frac{p(x)}{(ax + b)^n} dx, \text{ where } p(x) \text{ is a polynomial}$$

and n is a positive rational number

Steps:

- i. Substitute $ax + b = v$ or $x = \frac{v - b}{a}$ and $dx = \frac{1}{a} dv$
- ii. Now integrate with respect to v by using $\int v^n dv = \frac{v^{n+1}}{n+1} + C$
- iii. Replace v by $ax + b$

8. Evaluation of integrals of the form

$$\int \tan^m x \cdot \sec^{2n} x dx, \int \cot^m x \cdot \operatorname{cosec}^{2n} x dx, m, n \in \mathbb{N}$$

Steps:

i. Rewrite the given integral as $I = \int \tan^m x \cdot (\sec^2 x)^{(n-1)} \sec^2 x dx$

ii. Substitute $\tan x = v$ and $\sec^2 x dx = dv$

Therefore,

$$\begin{aligned} I &= \int \tan^m x \cdot (\sec^2 x)^{(n-1)} \sec^2 x dx \\ &= \int \tan^m x \cdot (1 + \tan^2 x)^{(n-1)} \sec^2 x dx \\ &= \int v^m \cdot (1 + v^2)^{(n-1)} dv \end{aligned}$$

iii. Use the binomial theorem to expand $(1 + v^2)^{(n-1)}$ in step (ii) and integrate

iv. Replace v by $\tan x$ in step (iii)

9. Evaluation of integrals of the form

$$\int \tan^{2m+1} x \cdot \sec^{2n+1} x dx, \int \cot^m x \cdot \operatorname{cosec}^{2n} x dx, \text{ where } m \text{ and } n \text{ are nonnegative integers}$$

Steps:

i. Rewrite the given integral as $I = \int (\tan^2 x)^m \cdot (\sec x)^{2n} \sec x \cdot \tan x dx$

ii. Put $\sec x = v$ and $\sec x \tan x dx = dv$

Therefore,

$$\begin{aligned} I &= \int (\sec^2 x - 1)^m \cdot (\sec x)^{2n} \sec x \cdot \tan x dx \\ &= \int (v^2 - 1)^m v^n dv \end{aligned}$$

iii. Use the binomial theorem to expand $(v^2 - 1)^m$ in step (ii) and integrate

iv. Replace v by $\sec x$ in step (iii)

10. Evaluation of integrals of the form $\int x \cdot \cos^n x dx$, where $m, n \in \mathbb{N}$

Steps:

i. Check the exponents of $\sin x$ and $\cos x$

ii. If the exponent of $\sin x$ is an odd positive integer, then put $\cos x = v$

If the exponent of $\cos x$ is an odd positive integer, then put $\sin x = v$

If the exponents of both $\sin x$ and $\cos x$ are odd positive integers, then put either $\sin x = v$ or $\cos x = v$

If the exponents of both $\sin x$ and $\cos x$ are even positive integers, then rewrite $\sin^m x \cos^n x$ in terms of sines and cosines of multiples of x by using trigonometric results.

iii. Evaluate the integral in step (ii)

11. Evaluation of integrals of the form

$\int \sin^m x \cdot \cos^n x dx$, where $m, n \in \mathbb{Q}$, such that $m+n$ is a negative even integer.

Steps:

- Represent the integrand in terms of $\tan x$ and $\sec^2 x$ by dividing the numerator and denominator by $\cos^k x$, where $k = -(m + n)$
- Put $\tan x = v$

12. Evaluation of integrals of the form $\int \frac{dx}{ax^2+bx+c}$

Steps:

- Multiply and divide the integrand by x^2 and make the coefficient of x^2 unity
- Observe the coefficient of x
- Add and subtract $\left(\frac{1}{2} \text{coefficient } x\right)^2$ to the expression in the denominator

$$\left\{ \left(x + \frac{b}{2a} \right)^2 + \frac{4ac - b^2}{4a^2} \right\}$$

- Express the expression in the denominator in the form
- Use the appropriate formula to integrate.

13. Evaluation of integrals of the form $\int \frac{dx}{ax^2+bx+c}$

Steps:

- Multiply and divide the integrand by x^2 and make the coefficient of x^2 unity
- Observe the coefficient of x
- Add and subtract $\left(\frac{1}{2} \text{coefficient } x\right)^2$ inside the square root

$$\left\{ \left(x + \frac{b}{2a} \right)^2 + \frac{4ac - b^2}{4a^2} \right\}$$

- Express the expression inside the square root in the form
- Use the appropriate formula to integrate.

14. Evaluation of integrals of the form $\int \frac{px+q}{ax^2+bx+c} dx$

Steps:

- Rewrite the numerator as follows:

$$px + q = A \left\{ \frac{d}{dx} (ax^2 + bx + c) \right\} + B$$

$$\Rightarrow px + q = A \{2ax + b\} + B$$

- ii. Find the values of A and B by equating the coefficients of like powers of x on both sides
- iii. Substitute $px+q$ by $A \{2ax + b\} + B$ in the given integral

$$\text{Therefore, } \int \frac{px + q}{ax^2 + bx + c} dx = A \int \frac{2ax + b}{ax^2 + bx + c} dx + B \int \frac{1}{ax^2 + bx + c} dx$$

- iv. Integrate the right-hand side and substitute the values of A and B

15. Evaluation of integrals of the form $\int \frac{px+q}{ax^2+bx+c} dx$ where $p(x)$ is a polynomial degree greater than or equal to 2

Steps:

- i. Divide the numerator by the denominator, and rewrite the integrand as

$$q(x) + \frac{r(x)}{ax^2 + bx + c}, \text{ where } r(x) \text{ is a linear function of } x$$

- ii. Thus,

$$\int \frac{px + q}{ax^2 + bx + c} dx = \int q(x) dx + \int \frac{r(x)}{ax^2 + bx + c} dx$$

- iii. Integrate the second integral on the right-hand side and apply the appropriate method

16. Evaluation of integrals of the form $\int \frac{px+q}{\sqrt{ax^2+bx+c}} dx$

Steps:

- i. Rewrite the numerator as follows:

$$px + q = A \left\{ \frac{d}{dx} (ax^2 + bx + c) \right\} + B$$

$$\Rightarrow px + q = A \{2ax + b\} + B$$

- ii. Find the values of A and B by equating the coefficients of like powers of x on both sides
- iii. Substitute $px + q$ by $A \{2ax + b\} + B$ in the given integral

$$\text{Therefore, } \int \frac{px + q}{\sqrt{ax^2 + bx + c}} dx = A \int \frac{2ax + b}{\sqrt{ax^2 + bx + c}} dx + B \int \frac{1}{\sqrt{ax^2 + bx + c}} dx$$

- iv. Integrate the right-hand side and substitute the values of A and B

17. Evaluation of integrals of the form

$$\int \frac{dx}{a \sin^2 x + b \cos^2 x}, \int \frac{dx}{a + b \sin^2 x}, \int \frac{dx}{a + b \cos^2 x},$$

$$\int \frac{dx}{(a \sin x + b \cos x)^2}, \int \frac{dx}{a + b \sin^2 x + c \cos^2 x}$$

Steps:

- i. Divide the numerator and denominator by $\cos^2 x$
- ii. In the denominator, replace $\sec^2 x$ by $1 + \tan^2 x$
- iii. Substitute $\tan x = v$; $\sec^2 x dx = dv$
- iv. Apply the appropriate method to integrate the integral $\int \frac{dv}{av^2 + bv + c}$

18. Evaluation of integrals of the form

$$\int \frac{dx}{a \sin x + b \cos x}, \int \frac{dx}{a + b \sin x}, \int \frac{dx}{a + b \cos x},$$

$$\int \frac{dx}{a + b \sin x + c \cos x}$$

Steps:

- i. Substitute $\sin x = \frac{2 \tan \frac{x}{2}}{1 + \tan^2 \frac{x}{2}}$, $\cos x = \frac{1 - \tan^2 \frac{x}{2}}{1 + \tan^2 \frac{x}{2}}$
- ii. In the numerator, replace $1 + \tan^2 \frac{x}{2}$ by $\sec^2 \frac{x}{2}$
- iii. Substitute $\tan \frac{x}{2} = v$; $\frac{1}{2} \sec^2 \frac{x}{2} dx = dv$
- iv. Apply the appropriate method to integrate the integral $\int \frac{dv}{av^2 + bv + c}$

19. Alternate method: Evaluation of integrals of the form

$$\int \frac{dx}{a \sin x + b \cos x}$$

Steps:

Substitute $a = r \cos \theta, b = r \sin \theta$;

where $r = \sqrt{a^2 + b^2}, \theta = \tan^{-1}\left(\frac{b}{a}\right)$

$$\Rightarrow a \sin x + b \cos x = r \cos \theta \sin x + r \sin \theta \cos x = r \sin(x + \theta)$$

$$\therefore \int \frac{dx}{a \sin x + b \cos x} = \frac{1}{\sqrt{a^2 + b^2}} \log \left| \tan \left(\frac{x}{2} + \frac{1}{2} \tan^{-1} \frac{b}{a} \right) \right| + C$$

20. Evaluation of integrals of the form

$$\int \frac{a \sin x + b \cos x}{c \sin x + d \cos x} dx$$

Steps:

i. Substitute

Numerator = A(Differentiation of denominator) + B(Denominator)

That is $a \sin x + b \cos x = A(c \cos x - d \sin x) + B(c \sin x + d \cos x)$

ii. Compare the coefficients of $\sin x$ and $\cos x$ on both the sides and get the values of A and B

iii. Replace the integrand by $A(c \cos x - d \sin x) + B(c \sin x + d \cos x)$

$$\begin{aligned} \text{iv. Hence, the value of the integral } \int \frac{a \sin x + b \cos x}{c \sin x + d \cos x} dx & \text{ is} \\ & = A \log |c \sin x + d \cos x| + \mu x + C \end{aligned}$$

21. Evaluation of integrals of the form

$$\int \frac{a \sin x + b \cos x + c}{m \sin x + n \cos x + p} dx$$

Steps:

i. Substitute

Numerator = A(Differentiation of denominator) + B(Denominator) + K

That is $a \sin x + b \cos x + c = A(m \cos x - n \sin x) + B(m \sin x + n \cos x + p) + K$

ii. Compare the coefficients of $\sin x$ and $\cos x$ and constant terms on both the sides and get the values of A, B and K

iii. Replace the integrand by $A(m \cos x - n \sin x) + B(m \sin x + n \cos x + p) + K$

$$\begin{aligned} \text{iv. Hence, the value of the integral } \int \frac{a \sin x + b \cos x + c}{m \sin x + n \cos x + p} dx & \text{ is} \\ & = A \log |m \sin x + n \cos x + p| + Bx + p \int \frac{1}{m \sin x + n \cos x + p} dx \end{aligned}$$

v. Evaluate the integral on the right-hand side by any appropriate method

22. Evaluation of integrals of the form

$$\int e^x [f(x) + f'(x)] dx$$

Steps:

- i. Write the given integral as

$$\int e^x [f(x) + f'(x)] dx = \int e^x f(x) dx + \int e^x f'(x) dx$$

- ii. Find the integration for the first term by parts

- iii. (Cancel out the second integral with the second term obtained by integration by parts)

- iv. Thus, the above result holds true for
- e^{kx}

$$\int e^{kx} [f(x) + f'(x)] dx = e^{kx} + C$$

23. Evaluation of the integrals of the form $\int \sqrt{ax^2 + bx + c} dx$

Steps:

- i. Take 'a' common inside the square root so as to get
- $x^2 + \frac{b}{a}x + \frac{c}{a}$

- ii. Add and subtract the appropriate term to
- $x^2 + \frac{b}{a}x + \frac{c}{a}$
- to get the term
- $\left(x + \frac{b}{2a}\right)^2 + \frac{4ac - b^2}{4a^2}$

- iii. Now evaluate the integral using the appropriate formulae

24. Evaluation of the integrals of the form $\int (mx + n)\sqrt{ax^2 + bx + c} dx$

Steps:

- i. Rewrite
- $mx + n$
- as
- $mx + n = A \frac{d}{dx}(ax^2 + bx + c) + B$

$$\text{That is, } mx + n = A(2ax + b) + B$$

- ii. Equate the coefficients of
- x
- and constant terms on both sides to get the values of
- A
- and
- B

- iii. Substitute
- $mx + n$
- by
- $A(2ax + b) + B$

- iv. Now evaluate the integral using the appropriate formulae

25. Evaluation of the integrals of the form

$$\int \frac{x^2 + 1}{x^4 + kx^2 + 1} dx, \int \frac{x^2 - 1}{x^4 + kx^2 + 1} dx, \int \frac{1}{x^4 + kx^2 + 1} dx \text{ where } k \in \mathbb{R}$$

Steps:

- i. Divide the numerator and the denominator by x^2
- ii. Write the denominator of the integrand in the form of $\left(x + \frac{1}{x}\right)^2 \pm m^2$
- iii. Write $d\left(x + \frac{1}{x}\right)$ or $d\left(x - \frac{1}{x}\right)$ or both in the numerator
- iv. Put $x + \frac{1}{x} = v$ or $x - \frac{1}{x} = v$
- v. Evaluate the integral using the appropriate formula

26. Evaluation of integration of irrational algebraic functions,

$$\int \frac{f(x)}{(gx + h)\sqrt{mx + n}} dx \text{ where } g, h, m, n \in \mathbb{R}$$

Step: Put $mx + n = v^2$ to evaluate the integral

27. Evaluation of integration of irrational algebraic functions,

$$\int \frac{f(x)}{(rx^2 + gx + h)\sqrt{mx + n}} dx \text{ where } r, g, h, m, n \in \mathbb{R}$$

Step: Put $mx + n = v^2$ to evaluate the integral

28. Evaluation of integration of irrational algebraic functions,

$$\int \frac{1}{(gx + h)\sqrt{mx^2 + nx + p}} dx \text{ where } g, h, m, n, p \in \mathbb{R}$$

Step: Put $gx + h = \frac{1}{v}$ to evaluate the integral

29. Evaluation of integration of irrational algebraic functions,

$$\int \frac{1}{(gx^2 + h)\sqrt{mx^2 + n}} dx \text{ where } g, h, m, n \in \mathbb{R}$$

Step:

i. Put $x = \frac{1}{v}$

$$\text{Therefore, } I = \int \frac{-v dv}{(g + hv^2)\sqrt{m + nv^2}}$$

$$\text{Now substitute } m + nv^2 = w^2$$

6. Properties of definite integrals

$$\bullet \int_a^b f(x) dx = \int_a^b f(t) dt$$

$$\bullet \int_a^b f(x) dx = - \int_b^a f(x) dx$$

$$\text{In particular, } \int_a^a f(x) dx = 0$$

$$\bullet \int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$$

$$\bullet \int_a^b f(x) dx = \int_a^b f(a + b - x) dx$$

$$\bullet \int_0^a f(x) dx = \int_0^a f(a - x) dx$$

$$\bullet \int_0^{2a} f(x) dx = \int_0^a f(x) dx + \int_0^a f(2a - x) dx$$

$$\bullet \int_0^{2a} f(x) dx = 2 \int_0^a f(x) dx, \text{ if } f(2a - x) = f(x)$$

$$= 0, \text{ if } f(2a - x) = -f(x)$$

$$\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx, \text{ if } f(-x) = f(x)$$

$$= 0, \text{ if } f(-x) = -f(x)$$

Class : 12th Maths
Chapter- 7 : Integrals

The method in which we change the variable to some other variable is called the method of substitution

$$\int \tan x dx = \log |\sec x| + c \quad \int \cot x dx = \log |\sin x| + c$$

$$\int \sec x dx = \log |\sec x + \tan x| + c \quad \int \operatorname{cosec} x dx = \log |\operatorname{cosec} x - \cot x| + c$$

$$(i) \int \frac{dx}{x^2 - a^2} = \frac{1}{2a} \log \left| \frac{x-a}{x+a} \right| + c \quad (ii) \int \frac{dx}{a^2 - x^2} = \frac{1}{2a} \log \left| \frac{a+x}{a-x} \right| + c$$

$$(iii) \int \frac{dx}{x^2 + a^2} = \frac{1}{a} \tan^{-1} \frac{x}{a} + c \quad (iv) \int \frac{dx}{\sqrt{x^2 - a^2}} = \log |x + \sqrt{x^2 - a^2}| + c$$

$$(v) \int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1} \frac{x}{a} + c \quad (vi) \int \frac{dx}{\sqrt{x^2 + a^2}} = \log |x + \sqrt{x^2 + a^2}| + c$$

It is the inverse of differentiation. Let, $\frac{d}{dx} F(x) = f(x)$. Then, $\int f(x) dx = F(x) + c$, 'c' : constant of integral. These integrals are called indefinite or general integrals.

Properties of indefinite integrals are

$$(i) \int [f(x) + g(x)] dx = \int f(x) dx + \int g(x) dx, \quad (ii) \int kf(x) dx = k \int f(x) dx,$$

For eg : $\int (3x^2 + 2x) dx = x^3 + x^2 + c$ where k is real.

Integration by substitution

Integration

integration of
some special
functions

Integrals

First fundamental
theorem of integral
calculus

Integration by parts

Second fundamental
theorem of integral Calculus

Some special type of
integrals

Example

Integration by
partial fractions

Let the area function be defined by:
 $A(x) = \int_a^x f(x) dx \forall x \geq a$,
where f is continuous on $[a, b]$
then $A'(x) = f(x) \forall x \in [a, b]$.

$$\int f_1(x) f_2(x) dx = f_1(x) \int f_2(x) dx - \int \left[\frac{d}{dx} f_1(x) \int f_2(x) dx \right] dx$$

Let f be a continuous function of x defined on $[a, b]$ and let F be another function such that $\frac{d}{dx} F(x) = f(x) \forall x \in \text{domain of } f$, then $\int_a^b f(x) dx = [F(x) + c]_a^b = F(b) - F(a)$. This is called the definite integral of f over the range $[a, b]$, where a and b are called the limits of integration, a being the lower limit and b be the upper limit.

A rational function of the form $\frac{P(x)}{Q(x)} (Q(x) \neq 0) = T(x) + \frac{P_1(x)}{Q(x)}$, $P_1(x)$

has degree less than that of $Q(x)$. We can integrate $\frac{P_1(x)}{Q(x)}$ by expressing

it in the following forms –

$$(i) \frac{px+q}{(x-a)(x-b)} = \frac{A}{x-a} + \frac{B}{x-b}, a \neq b.$$

$$(ii) \frac{px+q}{(x+a)^2} = \frac{A}{x-a} + \frac{B}{(x-a)^2}$$

$$(iii) \frac{px^2+qx+r}{(x-a)(x-b)(x-c)} = \frac{A}{x-a} + \frac{B}{x-b} + \frac{C}{x-c}$$

$$(iv) \frac{px^2+qx+r}{(x-a)^2(x-b)} = \frac{A}{x-a} + \frac{B}{(x-a)^2} + \frac{C}{x-b}$$

$$(v) \frac{px^2+qx+r}{(x-a)(x^2+bx+c)} = \frac{A}{x-a} + \frac{Bx+C}{x^2+bx+c}$$

$$(i) \int x^n dx = \frac{x^{n+1}}{n+1} + c, n \neq -1 \text{ like, } \int dx = x + c$$

$$(ii) \int \cos x dx = \sin x + c \quad (iii) \int \sin x dx = -\cos x + c$$

$$(iv) \int \sec^2 x dx = \tan x + c \quad (v) \int \operatorname{cosec}^2 x dx = -\cot x + c$$

$$(vi) \int \sec x \tan x dx = \sec x + c \quad (vii) \int \operatorname{cosec} x \cot x dx = -\operatorname{cosec} x + c$$

$$(viii) \int \frac{dx}{\sqrt{1-x^2}} = \sin^{-1} x + c \quad (ix) \int \frac{dx}{\sqrt{1-x^2}} = -\cos^{-1} x + c$$

$$(x) \int \frac{dx}{1+x^2} = \tan^{-1} x + c \quad (xi) \int \frac{dx}{1+x^2} = -\cot^{-1} x + c$$

$$(xii) \int e^x dx = e^x + c \quad (xiii) \int a^x dx = \frac{a^x}{\log a} + c$$

$$(xiv) \int \frac{dx}{x\sqrt{x^2-1}} = \sec^{-1} x + c \quad (xv) \int \frac{dx}{x\sqrt{x^2-1}} = -\operatorname{cosec}^{-1} x + c$$

$$(xvi) \int \frac{1}{x} dx = \log|x| + c$$

Some Standard
integrals

$$(i) \int \sqrt{x^2 - a^2} dx = \frac{x}{2} \sqrt{x^2 - a^2} - \frac{a^2}{2} \log |x + \sqrt{x^2 - a^2}| + c.$$

$$(ii) \int \sqrt{x^2 + a^2} dx = \frac{x}{2} \sqrt{x^2 + a^2} + \frac{a^2}{2} \log |x + \sqrt{x^2 + a^2}| + c$$

$$(iii) \int \sqrt{a^2 - x^2} dx = \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \frac{x}{a} + c.$$

$$\int_{-\pi/4}^{\pi/4} \sin^2 x dx$$

$$= 2 \int_{-\pi/4}^{\pi/4} \sin^2 x dx$$

$$= 2 \int_0^{\pi/4} \left(\frac{1 - \cos 2x}{2} \right) dx$$

$$= \int_0^{\pi/4} (1 - \cos 2x) dx$$

$$= \left[x - \frac{\sin 2x}{2} \right]_0^{\pi/4}$$

$$= \frac{\pi}{4} - \frac{1}{2}$$

Important Questions

Multiple Choice questions-

1. The anti-derivative of $(\sqrt{x} + \frac{1}{\sqrt{x}})$ equals

(a) $\frac{1}{3}x^{\frac{1}{3}} + 2x^{\frac{1}{2}} + c$

(b) $\frac{2}{3}x^{\frac{2}{3}} + \frac{1}{2}x^2 + c$

(c) $\frac{2}{3}x^{\frac{3}{2}} + 2x^{\frac{1}{2}} + c$

(d) $\frac{3}{2}x^{\frac{3}{2}} + \frac{1}{2}x^{\frac{1}{2}} + c$

2. If $\frac{1}{dx}(f(x)) = 4x^3 - \frac{3}{x^4}$ such that $f(2) = 0$ then $f(x)$ is

(a) $x^4 + \frac{1}{x^3} - \frac{129}{8}$

(b) $x^3 + \frac{1}{x^4} + \frac{129}{8}$

(c) $x^4 + \frac{1}{x^3} + \frac{129}{8}$

(d) $x^3 + \frac{1}{x^4} - \frac{129}{8}$

3. $\int \frac{10x^9 + 10^x \log_e 10}{x^{10} + 10^x} dx$ equals

(a) $10^x - x^{10} + c$

(b) $10^x + x^{10} + c$

(c) $(10^x - x^{10}) - 1 + c$

(d) $\log(10^x + x^{10}) + c$

4. $\int \frac{dx}{\sin^2 x \cos^2 x}$ equals

(a) $\tan x + \cot x + c$

(b) $\tan x - \cot x + c$

(c) $\tan x \cot x + c$

(d) $\tan x - \cot 2x + c$

5. $\int \frac{\sin^2 x - \cos^2 x}{\sin^2 x \cos^2 x} dx$ is equals to

(a) $\tan x + \cot x + c$

(b) $\tan x + \operatorname{cosec} x + c$

(c) $-\tan x + \cot x + c$

(d) $\tan x + \sec x + c$

6. $\int \frac{e^x(1+x)}{\cos^2(xe^x)} dx$ is equals to

(a) $-\cot(xe^x) + c$

(b) $\tan(xe^x) + c$

(c) $\tan(e^x) + c$

(d) $\cot(e^x) + c$

7. $\int \frac{dx}{x^2+2x+2}$ equals

(a) $x \tan^{-1}(x+1) + c$

(b) $\tan^{-1}(x+1) + c$

(c) $(x+1) \tan^{-1} x + c$

(d) $\tan^{-1} x + c$

8. $\int \frac{dx}{\sqrt{9-25x^2}}$ equals

(a) $\sin^{-1}\left(\frac{5x}{3}\right) + c$

(b) $\frac{1}{5} \sin^{-1}\left(\frac{5x}{3}\right) + c$

(c) $\frac{1}{6} \log\left(\frac{3+5x}{3-5x}\right) + c$

(d) $\frac{1}{30} \log\left(\frac{3+5x}{3-5x}\right) + c$

9. $\int \frac{x dx}{(x-1)(x-2)}$ equals

$$(a) \log \left| \left(\frac{x-1}{x-2} \right)^2 \right| + c$$

$$(b) \log \left| \frac{(x-2)^2}{x-1} \right| + c$$

$$(c) \log \left| \left(\frac{x-1}{x-2} \right)^2 \right| + c$$

$$(d) \log |(x-1)(x-2)| + c.$$

10. $\int \frac{dx}{x(x^2+1)}$ equals

$$(a) \log |x| - \frac{1}{2} \log (x^2 + 1) + c$$

$$(b) \frac{1}{2} \log |x| + \frac{1}{2} \log (x^2 + 1) + c$$

$$(c) -\log |x| + \frac{1}{2} \log (x^2 + 1) + c$$

$$(d) \log |x| + \log (x^2 + 1) + c$$

Very Short Questions:

1. Find $\int \frac{3+3\cos x}{x+\sin x} dx$ (C.B.S.E. Sample Paper 2019-20)

2. Find: $\int (\cos^2 2x - \sin^2 2x) dx$. (C.B.S.E. Sample Paper 2019-20)

3. Find: $\int \frac{dx}{\sqrt{5-4x-2x^2}}$ (C.B.S.E. Outside Delhi 2019)

4. Evaluate $\int \frac{x^3-1}{x^2} dx$ (N.C.E.R.T. C.B.S.E. 2010C)

5. Find: $\int \frac{\sin^2 x - \cos^2 x}{\sin x \cos x} dx$ (A.I.C.B.S.E. 2017)

6. Write the value of $\int \frac{dx}{x^2+16}$

7. Evaluate: $\int (x^3 + 1) dx$. (C.B.S.E. Sample Paper 2019-20)

8. Evaluate: $\int_0^{\pi/2} e^x (\sin x - \cos x) dx$. (C.B.S.E. 2014)

9. Evaluate: $\int_0^2 \sqrt{4-x^2} dx$ (A.I.C.B.S.E. 2014)

10. Evaluate: If $f(x) = \int_0^x t \sin t dt$, then write the value of $f'(x)$. (A.I. C.B.S.E. 2014)

E

Short Questions:

1. Evaluate:

$$\int \frac{\cos 2x + 2 \sin^2 x}{\cos^2 x} dx \text{ (C.B.S.E)}$$

2. Find: $\int \frac{\sec^2 x}{\sqrt{\tan^2 x + 4}} dx$

3. Find: $\int \sqrt{1 - \sin 2x} dx, \frac{\pi}{4} < x < \frac{\pi}{2}$

4. Find $\int \sin x \cdot \log \cos x dx$ (C.B.S.E 2019 C)

5. Find: $\int \frac{(x^2 + \sin^2 x) \sec^2 x}{1 + x^2} dx$ (CBSE Sample Paper 2018-19)

6. Evaluate $\int \frac{e^x(x-3)}{(x-1)^3} dx$ (CBSE Sample Paper 2018-19)

7. Find $\int \sin^{-1}(2x) dx$

8. Evaluate: $\int_{-\pi}^{\pi} (1 - x^2) \sin x \cos 2x dx$.

Long Questions:

1. Evaluate: $\int \frac{\sin^6 x + \cos^6 x}{\sin^2 x \cos^2 x} dx$ (C.B.S.E. 2019 (Delhi))

2. Integrate the function $\frac{\cos(x+a)}{\sin(x+b)}$ w.r.t. x. (C.B.S.E. 2019 (Delhi))

3. Evaluate: $\int x^2 \tan^{-1} x dx$. (C.B.S.E. (F) 2012)

4. Find: $\int \left[\log(\log x) + \frac{1}{(\log x)^2} \right] dx$ (N.C.E.R.T.; A.I.C.B.S.E. 2010 C)

Case Study Questions-

1. Integration is the process of finding the antiderivative of a function. In this process, we are provided with the derivative of a function and asked to find out the function (i.e., Primitive). Integration is the inverse process of differentiation.

Let $f(x)$ be a function of x . If there is a function $g(x)$, such that $d/dx (g(x)) = f(x)$, then $g(x)$ is called an integral of $f(x)$ w.r.t x and is denoted by $\int f(x) dx = g(x) + c$, where c is constant of integration.

(i) $\int (3x+4)^3 dx$ is equal to:

(a) $\frac{(3x+4)^4}{12} + c$

(b) $\frac{3(3x+4)^4}{4} + c$

(c) $\frac{3(3x+4)^2}{2} + c$

(d) $\frac{3(3x+4)^2}{4} + c$

(ii) $\int \frac{(x+1)^2}{x(x^2+1)} dx$ is equal to

(a) $\log|x| + c$

(b) $\log|x| + 2\tan^{-1}x + c$

(c) $-\log|x^2+1| + c$

(d) $\log|x(x^2+1)| + c$

(iii) $\int \sin^2(x) dx$ is equal to:

(a) $\frac{x}{2} + \frac{\sin 2x}{4} + c$

(b) $\frac{x}{2} - \frac{\sin 2x}{4} + c$

(c) $x + \frac{\sin 2x}{2} + c$

(d) $x - \frac{\sin 2x}{2} + c$

(iv) $\int \tan^2(x) dx$ is equal to:

(a) $\tan x + x + c$ (b) $-\tan x - x + c$

(c) $-\tan x + x + c$ (d) $\tan x - x + c$

(v) $\int \frac{dx}{\sin^2 x \cos^2 x}$ is equal to

(a) $2 \tan 2x + c$

(b) $-2 \tan 2x + c$

(c) $-2 \cot 2x + c$

(d) $2 \cot 2x + c$

2. When the integrated can be expressed as a product of two functions, one of which can be differentiated and the other can be integrated, then we apply integration by parts. If $f(x)$ = first function (that can be differentiated) and $g(x)$ = second function (that can be integrated), then the preference of this order can be decided by the word "ILATE", where

I stands for Inverse Trigonometric Function

L stands for Logarithmic Function

A stands for Algebraic Function

T stands for Trigonometric Function

E stands for Exponential Function, then

$$\int f(x)g(x)dx = f(x)\int g(x)dx - \int \left\{ \frac{d}{dx} f(x) \int g(x)dx \right\} dx$$

(i) $\int x \cdot \sin 3x dx =$

(a) $\frac{x \cos 3x}{3} - \frac{\sin 3x}{9} + c$

(b) $-\frac{x \cos 3x}{3} + \frac{\sin 3x}{9} + c$

(c) $\frac{x \cos 3x}{3} + \frac{\sin 3x}{9} + c$

(d) $-\frac{x \cos 3x}{3} - \frac{\sin 3x}{9} + c$

(ii) $\int \log(x+1) dx =$

(a) $\log(x+1) - x + c$

(b) $x \log(x+1) - x + c$

(c) $x \log(x+1) - \log(x+1) + x + c$

(d) $x \log(x+1) + \log(x+1) - x + c$

(iii) $\int x^2 e^{3x} dx =$

(a) $\frac{e^{3x}}{9} (9x^2 + 6x + 2) + c$

(b) $\frac{e^{3x}}{9} (9x^2 - 6x + 2) + c$

(c) $\frac{e^{3x}}{27} (9x^2 + 6x + 2) + c$

(d) $\frac{e^{3x}}{27} (9x^2 - 6x + 2) + c$

(iv) $\int (f(x)g''(x) - f''(x)g(x)) dx =$

(a) $f(x)g'(x) - f'(x)g(x) + c$

(b) $f(x)g'(x) + f'(x)g(x) + c$

(c) $f'(x)g(x) - f(x)g'(x) + c$

(d) $\frac{f(x)}{g'(x)} + c$

Answer Key-

Multiple Choice questions-

1. Answer: (c) $\frac{2}{3} x^{\frac{2}{3}} + 2x^{\frac{1}{2}} + c$

2. Answer: (a) $x^4 + \frac{1}{x^3} - \frac{129}{8}$

3. Answer: (d) $\log(10^x + x^{10}) + c.$

4. Answer: (b) $\tan x - \cot x + c$
5. Answer: (a) $\tan x + \cot x + c$
6. Answer: (b) $\tan (xe^x) + c$
7. Answer: (b) $\tan^{-1} (x + 1) + c$
8. Answer: (b) $\frac{1}{5} \sin^{-1} \left(\frac{5x}{3} \right) + c$
9. Answer: (b) $\log \left| \frac{(x-2)^2}{x-1} \right| + c$
10. Answer: (a) $\log |x| - \frac{1}{2} \log (x^2 + 1) + c$

Very Short Answer:

1. Solution:

$$I = \int \frac{3+3\cos x}{x+\sin x} dx = 3 \log |x + \sin x| + c.$$

$$[\because \text{Num.} = \frac{d}{dx} \text{denom.}]$$

2. Solution:

$$I = \int \cos 4x dx = \frac{\sin 4x}{4} + c$$

3. Solution:

$$\begin{aligned} I &= \int \frac{dx}{\sqrt{5-4x-2x^2}} \\ &= \frac{1}{\sqrt{2}} \int \frac{dx}{\sqrt{\frac{5}{2}-2x-x^2}} \\ &= \frac{1}{\sqrt{2}} \int \frac{dx}{\sqrt{\left(\frac{\sqrt{7}}{\sqrt{2}}\right)^2 - (x+1)^2}} \\ &= \frac{1}{\sqrt{2}} \sin^{-1} \left[\frac{\sqrt{2}}{\sqrt{7}} (x+1) \right] + C. \end{aligned}$$

4. Solution:

$$\begin{aligned}
 \int \frac{x^3 - 1}{x^2} dx &= \int \left(x - \frac{1}{x^2} \right) dx \\
 &= \int x dx - \int x^{-2} dx \\
 &= \frac{x^2}{2} - \frac{x^{-1}}{-1} + c \\
 &= \frac{x^2}{2} + \frac{1}{x} + c, x \neq 0.
 \end{aligned}$$

5. Solution:

$$\begin{aligned}
 I &= \int \frac{\sin^2 x - \cos^2 x}{\sin x \cos x} dx \\
 &= \int \frac{\sin^2 x}{\sin x \cos x} dx - \int \frac{\cos^2 x}{\sin x \cos x} dx \\
 &= \int \tan x dx - \int \cot x dx \\
 &= \log |\sec x| - \log |\sin x| + c.
 \end{aligned}$$

6. Solution:

$$\begin{aligned}
 \int \frac{dx}{x^2 + 16} &= \int \frac{dx}{4^2 + x^2} \\
 & \quad | \text{ "Form : } \int \frac{dx}{a^2 + x^2} " \\
 &= \frac{1}{4} \tan^{-1} \frac{x}{4} + c.
 \end{aligned}$$

7. Solution:

$$\begin{aligned}
 I &= \int_{-2}^2 x^3 dx + \int_{-2}^2 1 \cdot dx = I_1 \\
 &\Rightarrow 0 + [x]_{-2}^2 \quad [\because I_1 \text{ is an odd function}] = 2 - (-2) = 4. \\
 &\Rightarrow 2 - (-2) = 4.
 \end{aligned}$$

8. Solution:

$$\int_0^{\pi/2} e^x (\sin x - \cos x) dx$$

$$\int_0^{\pi/2} e^x (-\cos x + \sin x) dx$$

$$I'' \text{ Form: } \int e^x (f(x) + f'(x)) dx$$

$$= [e^x (-\cos x)]_0^{\pi/2}$$

$$= -e^{\pi/2} \cos \frac{\pi}{2} + e^0 \cos 0$$

$$= -e^{\pi/2} (0) + (1) (1)$$

$$= -0 + 1 = 1$$

9. Solution:

$$\int_0^2 \sqrt{4-x^2} dx = \int_0^2 \sqrt{2^2-x^2} dx$$

$$| \text{“Form : } \int \sqrt{a^2-x^2} dx”$$

$$= \left[\frac{x\sqrt{4-x^2}}{2} + \frac{4}{2} \sin^{-1} \frac{x}{2} \right]_0^2$$

$$= [0 + 2 \sin^{-1}(1)] - [0 + 0]$$

$$= 2 \sin^{-1}(1) = 2(\pi/2) = \pi$$

10.

olution:

$$\text{We have: } f(x) = \int_0^x t \sin t dt.$$

$$f'(x) = x \sin x \cdot \frac{d}{dx}(x) - 0$$

[Property XII; Leibnitz's Rule]

$$= x \sin x \cdot (1)$$

$$= x \sin x.$$

Short Answer:

S

1. Solution:

$$\begin{aligned}
 I &= \int \frac{\cos 2x + 2\sin^2 x}{\cos^2 x} dx \\
 &= \int \frac{(1 - 2\sin^2 x) + 2\sin^2 x}{\cos^2 x} dx \\
 &= \int \frac{1}{\cos^2 x} dx = \int \sec^2 x dx \\
 &= \tan x + c.
 \end{aligned}$$

2. Solution:

$$I = \frac{\sec^2 x}{\sqrt{\tan^2 x + 4}}$$

Put $\tan x = t$ so that $\sec^2 x dx = dt$.

$$\therefore I = \int \frac{dt}{\sqrt{t^2 + 2^2}}$$

$$= \log |t + \sqrt{t^2 + 4}| + C$$

$$= \log |\tan x + \sqrt{t^2 + 4}| + C$$

3. Solution:

$$\begin{aligned}
 I &= \int \sqrt{1 - \sin 2x} dx \\
 &= \int \sqrt{(\sin^2 x + \cos^2 x) - 2\sin x \cos x} dx \\
 &= \int \sqrt{(\sin x - \cos x)^2} dx = \int (\sin x - \cos x) dx \\
 &= -\cos x - \sin x + C.
 \end{aligned}$$

4. Solution:

$$\int \sin x \cdot \log \cos x dx$$

Put $\cos x = t$

so that $-\sin x dx = dt$

i.e., $\sin x dx = -dt$.

$$\therefore I = -\int \log t \cdot 1 dt$$

$$= -[\log t \cdot t - \int 1/t \cdot t \, dt]$$

[Integrating by parts]

$$= -[t \log t - t] + C = t(1 - \log t) + C$$

$$= \cos x (1 - \log (\cos x)) + C.$$

5. Solution:

$$\begin{aligned} I &= \int \frac{(x^2 + \sin^2 x) \sec^2 x}{1 + x^2} dx \\ &= \int \left\{ \frac{(1 + x^2) + (\sin^2 x - 1)}{1 + x^2} \right\} \sec^2 x \, dx \\ &= \int \left(1 - \frac{\cos^2 x}{1 + x^2} \right) \sec^2 x \, dx \\ &= \int \sec^2 x \, dx - \int \frac{1}{1 + x^2} dx \\ &= \tan x - \tan^{-1} x + c. \end{aligned}$$

6. Solution:

$$\begin{aligned} I &= \int \frac{e^x(x-3)}{(x-1)^3} dx \\ &= \int e^x \left[\frac{(x-1)-2}{(x-1)^3} \right] dx \\ &= \int e^x \left[\frac{1}{(x-1)^2} - \frac{2}{(x-1)^3} \right] dx \\ &\quad \left| \text{“Form : } \int e^x [f(x) + f'(x)] dx \right| \\ &= \frac{e^x}{(x-1)^2} + c. \end{aligned}$$

7. Solution:

$$I = \int \sin^{-1}(2x) dx = \int \sin^{-1}(2x) \cdot 1 dx$$

$$= \sin^{-1}(2x) \cdot x - \int \frac{1}{\sqrt{1-4x^2}} (2)(x) dx$$

[Integrating by Parts]

$$= x \sin^{-1}(2x) - \int \frac{2x}{\sqrt{1-4x^2}} dx$$

$$= x \sin^{-1}(2x) + \frac{1}{4} \int (1-4x^2)^{-1/2} (-8x) dx$$

$$= x \sin^{-1}(2x) + \frac{1}{4} \frac{(1-4x^2)^{1/2}}{1/2} + C$$

$$= x \sin^{-1}(2x) + \frac{1}{2} \sqrt{1-4x^2} + C.$$

8. Solution:

$$\text{Here, } f(x) = (1-x^2) \sin x \cos^2 x.$$

$$f(x) = (1-x^2) \sin(-x) \cos^2(-x)$$

$$= -(1-x^2) \sin x \cos^2 x$$

$$= -f(x)$$

$$\Rightarrow f \text{ is an odd function.}$$

$$\text{Hence, } I = 0.$$

Long Answer:

1. Solution:

$$\begin{aligned}
 & \int \frac{\sin^6 x + \cos^6 x}{\sin^2 x \cos^2 x} dx \\
 &= \int \frac{(\sin^2 x + \cos^2 x)^3 - 3\sin^2 x \cos^2 x (\sin^2 x + \cos^2 x)}{\sin^2 x \cos^2 x} dx \\
 & \quad [\because a^3 + b^3 = (a+b)^3 - 3ab(a+b)] \\
 &= \int \frac{(1)^3 - 3\sin^2 x \cos^2 x (1)}{\sin^2 x \cos^2 x} dx \\
 &= \int \frac{1 - 3\sin^2 x \cos^2 x}{\sin^2 x \cos^2 x} dx \\
 &= \int \left(\frac{1}{\sin^2 x \cos^2 x} - 3 \right) dx \\
 &= \int \left(\frac{\sin^2 x + \cos^2 x}{\sin^2 x \cos^2 x} - 3 \right) dx \\
 & \quad \text{(Note this step)} \\
 &= \int \left(\frac{1}{\cos^2 x} + \frac{1}{\sin^2 x} - 3 \right) dx \\
 &= \int \sec^2 x dx + \int \operatorname{cosec}^2 x dx - 3 \int 1 dx \\
 &= \tan x - \cot x - 3x + c.
 \end{aligned}$$

2. Solution:

$$\begin{aligned}
 I &= \int \frac{\cos(x+a)}{\sin(x+b)} dx \\
 &= \int \frac{\cos(x+b+a-b)}{\sin(x+b)} dx \\
 & \quad \cos(x+b) \cos(a-b) \\
 &= \int \frac{-\sin(x+b) \sin(a-b)}{\sin(x+b)} dx
 \end{aligned}$$

$$\begin{aligned}
 &= \cos(a-b) \int \frac{\cos(x+a)}{\sin(x+b)} dx - \sin(a-b) \int 1 \cdot dx \\
 &= \cos(a-b) \log |\sin(x+b)| - \sin(a-b) \cdot x + c.
 \end{aligned}$$

3. Solution:

$$\begin{aligned}
 \int x^2 \tan^{-1} x \, dx &= \int \tan^{-1} x \cdot x^2 \, dx \\
 &= \tan^{-1} x \cdot \frac{x^3}{3} - \int \frac{1}{1+x^2} \cdot \frac{x^3}{3} dx \\
 &\quad \text{(Integrating by parts)} \\
 &= \frac{x^3}{3} \tan^{-1} x - \frac{1}{3} \int \frac{x^3}{1+x^2} dx \\
 &= \frac{x^3}{3} \tan^{-1} x - \frac{1}{3} \int \left(x - \frac{x}{1+x^2} \right) dx \\
 &= \frac{x^3}{3} \tan^{-1} x - \frac{1}{3} \int x \, dx + \frac{1}{6} \int \frac{2x}{1+x^2} dx \\
 &= \frac{x^3}{3} \tan^{-1} x - \frac{1}{3} \cdot \frac{x^2}{2} + \frac{1}{6} \log |1+x^2| + c \\
 &\quad \left[\because \frac{d}{dx}(1+x^2) = 2x \right] \\
 &= \frac{1}{3} x^3 \tan^{-1} x - \frac{1}{6} x^2 + \frac{1}{6} \log(1+x^2) + c. \\
 &\quad \left[\because x^2 \geq 0 \Rightarrow 1+x^2 > 0 \therefore |1+x^2| = 1+x^2 \right]
 \end{aligned}$$

4. Solution:

$$\begin{aligned}
 \text{Let } \int \left[\log(\log x) + \frac{1}{(\log x)^2} \right] dx \\
 = \int \log(\log x) \, dx + \int \frac{1}{(\log x)^2} \, dx \dots\dots (1)
 \end{aligned}$$

$$\text{Let } I = I_1 + I_2$$

$$\text{Now } I_1 = \int \log(\log x) \, dx$$

$$= \int \log(\log x) \cdot 1 \, dx$$

$$= \log(\log x) \cdot x - \int \frac{1}{\log x \cdot x} x \cdot dx$$

(Integrating by parts)

$$= x \log(\log x) - \int \frac{1}{\log x} dx \dots\dots\dots (2)$$

$$\text{Let } I_1 = I_3 + I_4$$

$$\begin{aligned} \text{And } I_4 &= \int \frac{1}{\log x} dx \\ &= \int \frac{1}{\log x} \cdot 1 dx \\ &= \frac{1}{\log x} \cdot x - \int -\left(\frac{1}{\log x}\right)^2 \cdot \frac{1}{x} \cdot x dx \\ &\quad \text{(Integrating by parts)} \\ &= \frac{x}{\log x} + \int \frac{1}{(\log x)^2} dx. \end{aligned}$$

Putting in (2),

$$I_1 = x \log(x) - \frac{x}{\log x} - \int \frac{1}{(\log x)^2} dx$$

Putting in (1),

$$I = x \log(\log x)$$

$$\begin{aligned} & - \frac{x}{\log x} - \int \frac{1}{(\log x)^2} dx + \int \frac{1}{(\log x)^2} dx \\ &= x \log(\log x) - \frac{x}{\log x} + c \\ &= x \left(\log(\log x) - \frac{1}{\log x} \right) + c. \end{aligned}$$

Case Study Answers-

1.

$$(i) \quad (a) \quad \frac{(3x+4)^4}{12} + c$$

$$(ii) \quad (b) \quad \log|x| + 2 \tan^{-1}x + c$$

(iii) (b) $\frac{x}{2} - \frac{\sin 2x}{4} + c$

(iv) (d) $\tan x - x + c$

(v) (c) $-2 \cot 2x + c$

2.

(i) (b) $-\frac{x \cos 3x}{3} + \frac{\sin 3x}{9} + c$

(ii) (d) $x \log(x+1) + \log(x+1) - x + c$

(iii) (d) $\frac{e^{3x}}{27} (9x^2 - 6x + 2) + c$

(iv) (a) $f(x)g'(x) - f'(x)g(x) + c$

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